

二重积分计算 (1)

微积分学堂 2024.4

一、直角坐标下计算

例 1 已知 $f(x, y)$ 有二阶连续偏导, 且 $f(1, y) = f(x, 1) = 0$, $\iint_D f(x, y) dx dy = a$,

这里 $D = [0, 1; 0, 1]$, 求 $I = \iint_D xy f''_{xy}(x, y) dx dy$.

解 注意 $f_y(1, y) = f_x(x, 1) = 0$, 用分部积分法计算逐次积分.

$$\begin{aligned} I &= \int_0^1 x dx \int_0^1 y \frac{\partial f_x}{\partial y} dy = \int_0^1 x \left[y f_x \Big|_{y=0}^{y=1} - \int_0^1 f_x dy \right] dx \\ &= \int_0^1 x \left[f_x(x, 1) - \int_0^1 f_x dy \right] dx = - \int_0^1 \left(x \int_0^1 f_x dy \right) dx \\ (\text{换序}) &= - \int_0^1 dy \int_0^1 x f_x dx = - \int_0^1 \left(x f \Big|_{x=0}^{x=1} - \int_0^1 f dx \right) dy \\ &= - \int_0^1 (f(1, y) - \int_0^1 f dx) dy = \int_0^1 \left(\int_0^1 f dx \right) dy = a. \end{aligned}$$

注 先对 x 后对 y 积分完全类似.

例 2 设 D 是由摆线 $\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$ 的一拱与 ox 轴所围的区域, 求 $\iint_D y^2 dx dy$.

解 $I = \int_0^{2\pi a} dx \int_0^{y(x)} y^2 dy$ ($y = y(x)$ 是摆线对应的函数)

$$= \frac{1}{3} \int_0^{2\pi a} y^3(x) dx$$

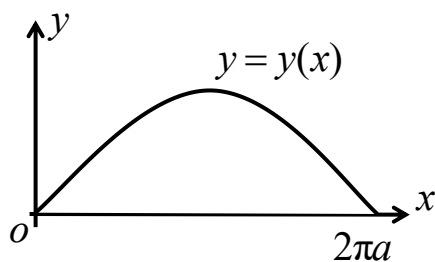
依据摆线的参数方程换元, 令 $x = a(t - \sin t)$, 则

$$y(x) = a(1 - \cos t).$$

$$I = \frac{1}{3} \int_0^{2\pi} (a(1 - \cos t))^3 \cdot a(1 - \cos t) dt$$

$$= \frac{a^4}{3} \int_0^{2\pi} (1 - \cos t)^4 dt = \frac{16a^4}{3} \int_0^{2\pi} \cos^8 \frac{t}{2} dt$$

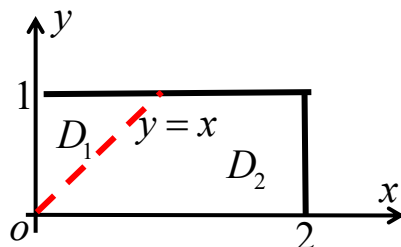
$$= \frac{32a^4}{3} \int_0^{\pi} \cos^8 \theta d\theta = \frac{32a^4}{3} \cdot 2 \cdot \left(\frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right) = \frac{35}{12} \pi a^4. \quad (\text{使用了 Wallis 公式})$$



例 3 求 $I = \iint_D \min\{x, y\} dx dy$, $D = [0, 2; 0, 1]$.

解 (区域可加性) 用直线 $y = x$ 分区域为 D_1, D_2 .

$$\begin{aligned} I &= \iint_{D_1} \min\{x, y\} dx dy + \iint_{D_2} \min\{x, y\} dx dy \\ &= \iint_{D_1} x dx dy + \iint_{D_2} y dx dy \\ &= \int_0^1 dy \int_0^y x dx + \int_0^1 dy \int_y^2 y dx \\ &= \frac{5}{6}. \end{aligned}$$



例 4 求 $I = \iint_D (|x| + y) dx dy$, D 由圆周 $x^2 + y^2 = 4$, 抛物线 $y = 1 - x^2$ ($|x| \leq 1$) 及

$y = x^2 - 1$ ($|x| \leq 1$) 所围.

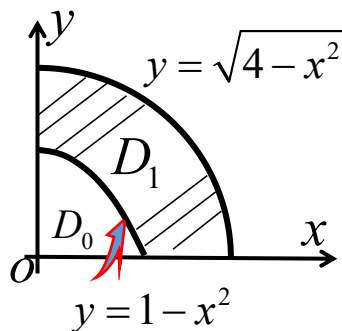
解 (用对称性, 偶倍奇零)

$$I = 4 \iint_{D_1} x dx dy.$$

而

$$\begin{aligned} \iint_{D_1} x dx dy &= \iint_{D_0 \cup D_1} x dx dy - \iint_{D_0} x dx dy \\ &= \int_0^2 dx \int_0^{\sqrt{4-x^2}} x dy - \int_0^1 dx \int_0^{1-x^2} x dy \\ &= \int_0^2 x \sqrt{4-x^2} dx - \int_0^1 x(1-x^2) dx = \frac{8}{3} - \frac{1}{4}, \end{aligned}$$

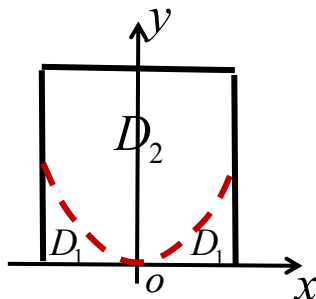
$$\text{故 } I = \frac{29}{3}.$$



例 5 求 $I = \iint_D |y - x^2| dx dy$, $D = [-1, 1; 0, 2]$.

解 (区域可加性) 用 $y = x^2$ 分区域为 D_1, D_2 .

$$\begin{aligned} I &= \iint_{D_1} |y - x^2| dx dy + \iint_{D_2} |y - x^2| dx dy \\ &= -\iint_{D_1} (y - x^2) dx dy + \iint_{D_2} (y - x^2) dx dy \\ &= -\int_{-1}^1 dx \int_0^{x^2} (y - x^2) dy + \int_{-1}^1 dx \int_{x^2}^2 (y - x^2) dy \\ &= \frac{46}{15}. \end{aligned}$$



例 6 求以圆盘 $D: x^2 + y^2 \leq R^2$ 为底, 以曲面 S :

$$(z-a)\varphi(x) + (z-b)\varphi(y) = 0$$

为顶的曲顶柱体的体积. 这里 $\varphi(t)$ 是正值连续函数, a, b 为正数.

解 曲面 S 可表示为 $z = \frac{a\varphi(x) + b\varphi(y)}{\varphi(x) + \varphi(y)}$, 曲顶柱体的体积为

$$\begin{aligned} V &= \iint_D \frac{a\varphi(x) + b\varphi(y)}{\varphi(x) + \varphi(y)} dx dy && \text{(用轮换对称性)} \\ &= \iint_D \frac{a\varphi(y) + b\varphi(x)}{\varphi(y) + \varphi(x)} dx dy \\ &= \frac{1}{2} \left\{ \iint_D \frac{a\varphi(x) + b\varphi(y)}{\varphi(x) + \varphi(y)} dx dy + \iint_D \frac{a\varphi(y) + b\varphi(x)}{\varphi(y) + \varphi(x)} dx dy \right\} \\ &= \frac{a+b}{2} \iint_D dx dy = \frac{a+b}{2} \pi R^2. \end{aligned}$$